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Cambridge IGCSE Combined and Co-ordinated Sciences

B1 Cells and Organisms

Comprehensive Exam Preparation Guide

B1 Cells and Organisms — Cambridge IGCSE Biology Study Guide

Chapter Overview

All learners study some content in this chapter.

In this chapter you will:

- Learn about the seven characteristics of living organisms
- Find out about the structure of the cells of bacteria, plants and animals
- Learn about the functions of each of the cells of bacteria, plants and animals
- Identify cell structures in diagrams
- Describe how the structures of some specialised cells are related to their functions
- State that new cells are produced by division of existing cells
- Describe the meaning of the terms cell, tissue, organ, organ system and organism
- Practise using the magnification equation
- Convert measurements between millimetres (mm) and micrometres (μm)

Important Terms

Term	Definition
Organism	A living thing, such as a human, a platypus, a bacterium or a mango tree
Cell	The smallest unit from which all organisms are made
Unicellular	Made of just a single cell; bacteria and yeast are unicellular
Multicellular	Made of many cells; humans and plants are multicellular
Movement	An action by an organism or part of an organism causing a change of position or place
Respiration	The chemical reactions in cells that break down glucose and other nutrient molecules to release energy for the cell to use
Sensitivity	The ability to detect changes in the internal or external environment and respond to them
Growth	A permanent increase in size and dry mass, by the growth of cells and the addition of new cells
Reproduction	The processes that make new organisms of the same species
Excretion	The removal from the body of the waste products of metabolism (unwanted or toxic substances) and of substances in excess of requirements
Nutrition	Taking in substances from the environment and using them for energy, growth and development
Metabolism	All the chemical reactions that take place in an organism's cells, including respiration

Metabolic reactions	Chemical reactions that take place in living organisms
Light microscope	A microscope that shines light through the specimen and uses glass lenses to magnify and focus the image; a very good one magnifies about ×1500
Electron microscope	A microscope that can magnify far more than a light microscope; needed to see ribosomes, and to see mitochondria clearly
Photomicrograph	A photograph taken using a light microscope
Electron micrograph	A picture taken using an electron microscope, such as Figure B1.10
Cell membrane	A very thin layer of protein and fat surrounding the cytoplasm of every cell; it controls what enters and leaves the cell
Partially permeable	Allows some molecules and ions to pass through, but not others
Cell wall	A tough layer outside the cell membrane; found in the cells of plants, fungi and bacteria
Cellulose	A polysaccharide that forms criss-crossing fibres; plant cell walls are made mainly of cellulose
Fully permeable	Allows all molecules and ions to pass through it
Cytoplasm	The jelly-like material that fills a cell; it is nearly all water, with many substances (especially proteins) dissolved in it
Vacuole	A fluid-filled space inside a cell, separated from the cytoplasm by a membrane
Cell sap	The fluid that fills the large vacuoles in plant cells; a solution of sugars and other substances
Vesicle	A very small vacuole
Nucleus	A structure containing DNA in the form of chromosomes (plural: nuclei)
Chromosome	A length of DNA, found in the nucleus of a cell; it contains genetic information in the form of many different genes
DNA	A molecule that contains genetic information, in the form of genes, that controls the proteins that are made in the cell
Chloroplasts	Small structures found inside some plant cells, inside which photosynthesis takes place
Chlorophyll	The green pigment inside chloroplasts that absorbs energy from sunlight for photosynthesis
Starch grains	Structures that consist of thousands of starch molecules, stored in some plant cells
Ribosome	A very small structure in a cell that uses information on DNA to make protein molecules
Mitochondrion	A small structure in a cell where aerobic respiration releases energy from glucose (plural: mitochondria)

Aerobic respiration	Chemical reactions in mitochondria that use oxygen to break down glucose and other nutrient molecules to release energy for the cell to use
Bacteria	Unicellular organisms whose cells do not contain a nucleus (singular: bacterium)
Prokaryotic cells	Cells with no nucleus; bacteria have prokaryotic cells
Circular DNA	The single large circle of DNA found in a bacterial cell instead of chromosomes in a nucleus; sometimes called a bacterial chromosome
Plasmids	Small, circular molecules of DNA found in many prokaryotic cells in addition to the main, much larger circle of DNA
Peptidoglycan	The substance that bacterial cell walls are made of, instead of cellulose
Specialised cell	A cell whose structure is modified to help it carry out a particular function effectively
Tissue	A group of similar cells that work together to perform a particular function
Epidermis	The outer layer of tissue on a plant; also the outer layer of an animal's skin
Organ	A structure made up of a group of tissues, working together to perform specific functions
Organ system	A group of organs with related functions, working together to perform body functions
Digestion	The breakdown of food
Magnification	How many times larger an image is than the actual object
Micrometre (μm)	A unit of length equal to one thousandth of a millimetre; 1 mm = 1000 μm and 1 μm = 1×10^{-6} m

B1.01 Characteristics of living organisms

Before You Start

The chapter begins with two questions to discuss with a partner.

Q: The list below contains some features of living organisms: breathing, excretion, a blood system, a nervous system, sensitivity, growth, reproduction, movement, nutrition, respiration. Which of these features are found in all living organisms?

A: Excretion, sensitivity, growth, reproduction, movement, nutrition and respiration. These are the seven characteristics of living things. Breathing, a blood system and a nervous system are not found in all organisms: plants and bacteria do not breathe (breathing is not the same as respiration), and they have no blood system and no nervous system.

Q: If you have a microscope, how can you distinguish between a cell from a plant, and a cell from an animal?

A: Look for a cell wall. All plant cells have a cell wall outside the cell membrane, but animal cells never do. You could also look for chloroplasts and a large central vacuole: if a cell has chloroplasts, it is a plant cell. However, a cell without chloroplasts could still be a plant cell (for example, a root cell), so the cell wall is the most reliable clue.

What Is an Organism?

Biology is the study of **organisms**. An organism is a complete living thing, such as yourself, a platypus, a bacterium or a mango tree. There are very many different kinds of organism on Earth, but all of them

share **seven characteristics** (Figure B1.01).

Some non-living things have some of these characteristics, but no non-living thing has all of them. For example, a car moves, takes in fuel and gives out waste gases, but it cannot grow or reproduce.

The Seven Characteristics of Living Organisms

A useful way to remember the seven characteristics is **MRS GREN**: Movement, Respiration, Sensitivity, Growth, Reproduction, Excretion, Nutrition.

Characteristic	What it means	Example
Movement	All organisms are able to move to some extent. Most animals can move their whole body from place to place, and plants can slowly move parts of themselves	A bird flying; a plant's leaves slowly turning towards the light
Respiration	All organisms break down glucose and other substances inside their cells, to release energy that they can use. This energy is used to make other chemical reactions in their cells happen	Every living cell respires all the time, day and night
Sensitivity	All organisms pick up information about changes in their environment, and react to the changes. The changes may be in the internal environment (such as the temperature of the blood) or the external environment (such as the intensity of sunlight)	You pull your hand away from something hot; a plant shoot grows towards the light
Growth	All organisms begin small and get larger, by the growth of their cells and by adding new cells to their bodies. Growth is a permanent increase in size or dry mass	A young kangaroo (joey) growing into an adult
Reproduction	Organisms are able to make new organisms of the same species as themselves	A kangaroo producing a joey; a mango tree producing seeds
Excretion	All organisms produce unwanted or toxic waste products as a result of their metabolic reactions, and these must be removed from the body	Breathing out carbon dioxide made by respiration; getting rid of urea in urine
Nutrition	Organisms take substances from their environment and use them to provide energy or materials to make new cells	A kangaroo eating plants; a plant making its own food by photosynthesis

Metabolism

The energy released by respiration is used to make the other chemical reactions in cells happen. All of these chemical reactions together, including respiration, are called **metabolism**. Each individual reaction is a **metabolic reaction**. Metabolism produces waste products, some of them toxic, which is why every organism needs to excrete.

Measuring Growth: Why Dry Mass?

Growth can be defined as a **permanent** increase in size or dry mass. In exam answers, use the Cambridge syllabus wording: growth is a permanent increase in size and dry mass. The word *permanent* matters: if you drink a glass of water you become heavier for a while, but that is not growth.

Measuring **dry mass** involves killing and drying the organism, then finding the mass of its body without any water. Because the organism has to be killed, this method is used more often for plants than for

animals. Dry mass is a more reliable measure of growth than ordinary (fresh) mass, because the amount of water in an organism goes up and down all the time.

What Figure B1.01 Shows

Figure B1.01 is a scene of red kangaroos in the Australian outback: an adult kangaroo with a joey in its pouch, a second kangaroo bending down to feed on plants, a bird flying overhead and trees in the background. Each of the seven characteristics is printed around the picture with a line pointing to part of the scene. For example, the flying bird shows movement, the joey in the pouch shows reproduction, and the feeding kangaroo shows nutrition. The message is that every living thing in the scene, including the trees, carries out all seven characteristics.

Science in Context B1.01: How Many Cells in a Human Body?

In some organisms, it is possible to count the number of cells in their bodies. In organisms such as humans, we can only estimate the number, because it would be impossible to count them all. Cells also differ in their volume, mass and how closely or loosely they are packed together, so you cannot simply divide body mass by the mass of one cell.

Scientists recently calculated the numbers of different cells and cell types in different organs, and added them up. The latest estimate suggests that the body of an adult human contains around **37 trillion cells**, which is 37 000 000 000 000.

Q: Can you think of any advantages for larger organisms of consisting of multiple cells, rather than just one larger cell?

A: In a multicellular organism, different cells can become specialised to carry out particular functions (for example, nerve cells, muscle cells and red blood cells), so the organism can do many different jobs efficiently. Other advantages: if some cells are damaged or die, they can be replaced and the organism survives; and small cells can exchange substances such as oxygen with their surroundings quickly, whereas in one giant cell substances would take too long to reach the middle.

Activity B1.01: Matching the Characteristics of Living Things with Their Descriptions

Aim: To learn the seven characteristics of living things and what each one means, by playing a memory card game.

Group size: A group of four or five.

You will need: 14 pieces of blank card, all exactly the same.

Method:

1. Write the seven characteristics of living things on seven of the pieces of card.
2. Write descriptions of each of the seven characteristics on the other seven cards.
3. Shuffle each set of cards. Place them face down in two rows of seven.
4. One person then selects a card from each row and turns them face up. If the name and description match, this person keeps the two cards. If they do not match, they place them face down again in the same positions.
5. Now the next person does the same.
6. Keep taking turns until all the cards have been taken by someone. The winner is the person with most cards at the end.

Expected results: Every name card is eventually paired with its correct description. Players who remember where cards are, and who know the descriptions well, collect the most pairs.

Conclusion: By the end you should be able to name all seven characteristics and match each one to its meaning. The cards must be identical so that nobody can recognise a card from its back.

Safety: There are no special hazards in this activity.

Questions B1.01 and B1.02 on this topic are answered in the Textbook Questions with Answers section.

B1.02 Cell structure

Organisms Are Made of Cells

All organisms are made of **cells**. Cells are very small, so large organisms contain millions of cells. Organisms made of many cells are **multicellular**. Some organisms are **unicellular**, which means that they are made of just a single cell. Bacteria and yeast are examples of single-celled organisms.

New Cells Come from Existing Cells

All cells are made from existing cells. New cells are formed when a fully grown cell **divides** into two. Cells never appear from non-living material. You can read about how cells divide in Chapter B12.

Microscopes

To see cells clearly, you need to use a microscope (Figure B1.02). The kind of microscope used in a school laboratory is called a **light microscope**, because it shines light through the piece of animal or plant you are looking at. It uses glass lenses to magnify and focus the image. A very good light microscope can magnify about **1500 times**.

Part of the light microscope (Figure B1.02)	What it does
Eyepiece	The lens at the top that you look through
Objective lenses	The lenses just above the specimen that magnify it; there are usually several, each giving a different magnification
Focusing knob	Turned to move the lenses and stage closer together or further apart until the image is sharp
Stage	Where the specimen is placed (usually on a glass slide)
Mirror	Reflects light up through the specimen

A photograph taken using a light microscope is called a **photomicrograph**. Some structures are too small to see clearly with a light microscope, so biologists use an **electron microscope**, which can magnify far more. A picture taken with an electron microscope is an **electron micrograph**.

Animal and Plant Cells Under a Light Microscope

Figure B1.03 (an animal cell): a cell with an irregular, rounded outline. It shows a thin cell membrane, cytoplasm, a large dark nucleus, several mitochondria, tiny dots labelled ribosomes, and a small vacuole or vesicle.

Figure B1.04 (a plant cell): a box-shaped cell. It shows a cell wall on the outside with the cell membrane pressed against it, a thin layer of cytoplasm around the edge containing the nucleus, green chloroplasts, mitochondria and ribosomes, and a very large vacuole filling most of the middle of the cell.

Ribosomes and mitochondria are drawn in these figures so that you know where they are. However, the book explains later that ribosomes are so small they can only be seen with an electron microscope, and mitochondria cannot be seen clearly without one.

Figure B1.05: a photomicrograph of cells from the trachea (windpipe) of a mammal. The cells have been stained (coloured) with a dye that makes the nucleus look darker than the cytoplasm. The cell membrane, cytoplasm and nucleus are labelled.

Figure B1.06: a photomicrograph of cells from a moss plant. The cell walls form a clear network, and each cell contains many green chloroplasts and a nucleus. You cannot see the cell membranes,

because they are pressed tightly against the inside of the cell walls.

Cell Membrane

Every cell has a **cell membrane** around the outside. The cell membrane is a very thin layer of protein and fat. It is very important to the cell because it **controls what goes in and out** of it.

The cell membrane is **partially permeable**. This means that it will let some substances through but not others. The cell membrane separates the contents of the cell from its environment (surroundings).

It is difficult to see the cell membrane in a plant cell, because it is pressed tightly against the inside of the cell wall.

Cell Wall

All plant cells are surrounded by a **cell wall** made mainly of **cellulose**. Animal cells never have cell walls. Cellulose belongs to a group of substances called polysaccharides (Chapter B3).

Cellulose forms fibres which criss-cross over one another to form a very strong covering to the cell. Figure B1.07 is a three-dimensional photograph of a cell wall surface, showing these criss-crossing cellulose fibres. The cell wall:

- Helps to protect and support the cell
- Prevents the cell from bursting if it absorbs a lot of water and swells

Because of the spaces between the fibres, even very large molecules are able to go through the cellulose cell wall. The cell wall is **fully permeable**, so it does not control what enters the cell; that is the job of the cell membrane.

Cell walls are also found in the cells of fungi and bacteria, but these walls are not made of cellulose.

Cytoplasm

Cytoplasm is a clear jelly. It is nearly all water, and it contains many substances dissolved in it, especially proteins. Many different **metabolic reactions** (the chemical reactions of life) take place in the cytoplasm.

Vacuole

A **vacuole** is a fluid-filled space inside a cell, surrounded by its own membrane.

- **Plant cells** have very large, **permanent vacuoles**, which contain a solution of sugars and other substances called **cell sap**. When the vacuole is full, it presses outwards on the rest of the cell and helps it to keep its shape.
- **Animal cells** have much smaller vacuoles, called **vesicles**, which also contain solutions.

Nucleus

The **nucleus** (plural: nuclei) is where the genetic information is stored. The information is kept on the **chromosomes**, which are inherited from the organism's parents. The chromosomes are made of **DNA**. The information carried on the DNA determines the kinds of proteins that are made in the cell.

Figure B1.08 is a photomicrograph of some plant cells dividing. The nuclei have been coloured with a blue stain, and the things that look like spider legs are chromosomes. We can normally only see chromosomes when a cell is dividing.

Chloroplasts

Chloroplasts are never found in animal cells, but most of the cells in the green parts of plants have them. They contain the green colouring, or pigment, called **chlorophyll**. Chlorophyll absorbs energy from sunlight, and this energy is then used to make food for the plant by **photosynthesis** (Chapter B5).

Chloroplasts often contain **starch grains**, which have been made by photosynthesis. Animal cells never contain starch grains.

Figure B1.09 shows a chloroplast from a cell in a pea plant. The large blue structure inside the chloroplast is a starch grain. The black stripes inside the chloroplast are membranes where the chlorophyll is kept.

Ribosomes

Ribosomes are tiny structures found in almost all animal cells and plant cells. They are so small that we can only see them with an electron microscope.

Ribosomes are where the cell **makes proteins**. The instructions on the DNA molecules are used to link together long chains of amino acids in a particular sequence. You can read more about protein molecules in Chapter B4, and more about how a cell makes them in Chapter B6.

Mitochondria

All the structures above (cell membrane, cell wall, cytoplasm, vacuoles, nucleus and chloroplasts) can be seen with a good light microscope. But some smaller structures cannot be seen clearly unless we use an electron microscope. These include **mitochondria** (singular: mitochondrion).

Mitochondria are found in almost all plant and animal cells. They are the parts of the cell where **aerobic respiration** happens. This is how energy is released from glucose (Chapter B9).

Aerobic respiration is the main way in which cells get the energy that they need to stay alive. The more energy a cell needs, the more mitochondria it has. Muscle cells, for example, are packed full of mitochondria.

Figure B1.10 is an electron micrograph of a mitochondrion, magnified 20 000 times. It is a long, oval structure. The pink lines are membranes, which separate the inside of the mitochondrion from the rest of the cell; the inner membrane makes many folds across the inside.

Figure B1.11 is a drawing of a mitochondrion cut open, showing it in three dimensions. Two membranes separate the inside of the mitochondrion from the rest of the cell, and the inner membrane is folded.

Summary Table: Structures in Animal and Plant Cells

Structure	Function	Animal cell	Plant cell
Cell membrane	Partially permeable; controls what enters and leaves the cell	Yes	Yes
Cell wall	Fully permeable; protects and supports the cell; stops it bursting when it takes up water	Never	Yes (made of cellulose)
Cytoplasm	Jelly where many metabolic reactions take place	Yes	Yes
Nucleus	Contains DNA in chromosomes; the DNA determines which proteins the cell makes	Yes	Yes
Vacuole	Fluid-filled space; in plants, the large vacuole stores cell sap and helps the cell keep its shape	Only small vacuoles (vesicles)	Large permanent vacuole containing cell sap
Chloroplasts	Absorb energy from sunlight for photosynthesis	Never	In most cells of the green parts
Starch grains	Store starch made by photosynthesis	Never	Sometimes (often inside chloroplasts)
Ribosomes	Make proteins	Yes	Yes

Mitochondria	Aerobic respiration, releasing energy from glucose	Yes	Yes

Key idea: plant cells have everything an animal cell has, plus a cell wall, a large permanent vacuole and (often) chloroplasts.

Bacterial Cells

Bacteria (singular: bacterium) are unicellular organisms. Bacterial cells are rather different from the cells of animals and plants. Figure B1.12 is a diagram of a bacterium: a capsule-shaped cell with labels for the cell wall (made of peptidoglycan, not cellulose), cell membrane, cytoplasm, ribosomes, circular DNA and plasmids.

No nucleus. The most important difference between a bacterial cell and animal or plant cells is that bacteria do not have a nucleus. Bacterial cells are also known as **prokaryotic cells**. 'Pro' means 'before', and 'karyotic' means 'nucleus'. Prokaryotic cells appeared on Earth millions of years before cells with nuclei appeared.

Circular DNA. Instead of chromosomes inside a nucleus, bacteria have a circle of DNA, sometimes called a bacterial chromosome. The DNA has exactly the same function as in other cells: it provides instructions for making proteins.

Plasmids. Bacterial cells often have one or more smaller circles of DNA, called **plasmids**. Scientists can use plasmids in the genetic modification of cells and organisms.

Cell wall. Bacterial cells always have a cell wall. Unlike plant cells, this cell wall is not made of cellulose. But its function is the same as in plant cells: it helps to support the cell, and prevents it from bursting if the cell takes up a lot of water.

Cell membrane. A partially permeable cell membrane is pressed tightly against the inside of the bacterial cell wall. As in plant and animal cells, the cell membrane controls what enters and leaves the cell.

Cytoplasm and ribosomes. Bacterial cells have cytoplasm and ribosomes, with the same functions as in animal and plant cells.

Missing structures. Bacterial cells do not have mitochondria or chloroplasts.

Identifying Cell Structures in Diagrams and Micrographs

- **Cell wall:** a thick outer layer, often drawn as a double line, around plant and bacterial cells. An animal cell has only a thin single line (its membrane) around it.
- **Cell membrane:** a thin line around the cytoplasm. In plant cells it lies right against the inside of the wall, so it is hard to see.
- **Nucleus:** a large round or oval body, usually the darkest part of a stained cell.
- **Large vacuole:** a big, clear space in the middle of a plant cell that pushes the cytoplasm and nucleus out to the edges.
- **Chloroplasts:** green oval bodies in the cytoplasm of plant cells.
- **Mitochondria:** small, sausage-shaped bodies. In electron micrographs, look for a double membrane with a folded inner membrane that looks like stripes.
- **Ribosomes:** tiny dots in the cytoplasm, seen only in electron micrographs.
- **Vesicles:** small round spaces surrounded by a membrane.
- **Bacterial cell:** no nucleus; a loop of DNA lies in the cytoplasm, often with small circles (plasmids).

Activity B1.02: Comparing Animal Cells and Plant Cells

Aim: To make a display that compares the structures of animal cells and plant cells.

Group size: A group of three or four.

Method summary: Decide how you will present your comparison. You could use annotated drawings, construct a large comparison table, or make a presentation. Use the information in this chapter, and you might also look for more pictures on the internet.

Expected results: A good display shows the **similarities** (both have a cell membrane, cytoplasm, a nucleus, ribosomes and mitochondria) and the **differences** (only plant cells have a cellulose cell wall, a large permanent vacuole containing cell sap, chloroplasts and starch grains; animal cells have only small vacuoles called vesicles). Annotations should give the function of every structure.

Conclusion: Plant and animal cells share the same basic structures, but plant cells have three extra features linked to their way of life: a cell wall for support, a large vacuole that keeps the cell firm, and chloroplasts for photosynthesis.

Safety: There are no laboratory hazards. When searching online, use reliable websites and check that pictures are labelled correctly.

Reflection: How Will You Learn the Parts of Cells?

The book suggests several ways to learn the names of the parts of animal, plant and bacterial cells, and their functions:

- Looking at diagrams and reading about the structures
- Practising drawing your own diagrams and labelling them
- Getting a friend to test you by asking questions
- Making some revision cards for yourself, with the name of a structure on one side and its function on the other side

Q: What other ideas might you try?

A: For example: draw a cell from memory and then check it against the book; use a mnemonic or colour code (for instance, highlight the three plant-only structures in green); teach the structures to someone else; make a three-column table (animal, plant, bacterial) and fill it in from memory; or practise labelling unlabelled diagrams and micrographs from past papers.

Question B1.07 (comparing bacterial, animal and plant cells) is answered with a full comparison table in the Textbook Questions with Answers section.

B1.03 Specialised cells

What Does Specialised Mean?

Multicellular organisms, such as humans or plants, may contain vast numbers of cells, and not all of these cells are alike. In a human body almost all of our cells have the same features that are found in most animal cells: a cell membrane, cytoplasm and a nucleus. But most of our cells have a particular function to perform, and their structure is modified to help them carry out that function effectively. They are **specialised**.

The same is true in a flowering plant: all the cells have the basic characteristics of plant cells, but they have slightly different structures that relate to their specific functions.

Table B1.01: Examples of Specialised Cells

Table B1.01 lists seven specialised cells (five from animals and two from plants) and where you can find out more about each one.

Specialised cell	Where it is found	Function	Where you can find out more
Root hair cell	Near the tips of the roots of flowering plants	Absorption of water and mineral ions	Chapter B2

Palisade mesophyll cell	In the leaves of flowering plants	Photosynthesis	Chapter B5
Red blood cell	In the blood of mammals	Transporting oxygen	Chapter B7
Ciliated cell	Lining the trachea and bronchi of animals	Cilia move mucus upwards	Chapter B9
Neurone	In the nervous system of animals	Conducting electrical impulses	Chapter B10
Sperm cell	Produced in the testes of mammals	The male gamete in sexual reproduction	Chapter B11
Egg cell	Produced in the ovaries of mammals	The female gamete in reproduction	Chapter B11

How Structure Is Related to Function

You will study each of these cells in later chapters. For now, notice how each cell's structure suits its job:

- **Root hair cell:** has a long, thin extension (the root hair), which gives a large surface area for absorbing water and mineral ions from the soil.
- **Palisade mesophyll cell:** is packed with chloroplasts, so it can absorb a lot of light energy for photosynthesis.
- **Red blood cell:** is full of haemoglobin, which carries oxygen; it has no nucleus, leaving more room for haemoglobin.
- **Ciliated cell:** has tiny hair-like cilia on its surface that beat to sweep mucus (with trapped dust and bacteria) up the trachea and bronchi.
- **Neurone:** has a very long, thin fibre that carries electrical impulses over long distances around the body.
- **Sperm cell:** has a tail for swimming to the egg, and many mitochondria to release the energy needed for swimming.
- **Egg cell:** is large and contains a store of food for the embryo after fertilisation.

Tissues

Often, cells that specialise in the same activity are found together. A group of cells like this is called a **tissue**.

An example of a tissue is the layer of cells lining your stomach. These cells make enzymes to help digest your food. **Figure B1.13** shows a row of these stomach-lining cells sitting side by side on a basement membrane. Each cell has a cell membrane, cytoplasm and a nucleus, plus vacuoles containing digestive enzymes near its top. Enzymes are shown being secreted out of the cells into the stomach.

The stomach also contains other tissues. For example, there is a layer of muscle in the stomach wall, made of cells which can move. This **muscle tissue** makes the wall of the stomach move in and out, churning the food and mixing it up with enzymes.

Plants also have tissues. You may already have looked at some **epidermis** tissue from an onion bulb. Inside a leaf, a layer of cells makes up the **palisade tissue**. These cells are specialised to carry out photosynthesis.

Organs

A group of different tissues that carry out a function together is called an **organ**. The stomach is an organ, made of tissues such as the lining tissue and the muscle tissue. Other organs include the heart,

the kidneys and the lungs.

In a plant, an onion bulb is an organ. A leaf is another example of a plant organ.

Organ Systems

The stomach is only one of the organs which help in the **digestion** (breakdown) of food. The mouth, the stomach and the intestines are all part of an **organ system** called the **digestive system**. Other examples:

- The heart is part of the **circulatory system**
- Each kidney is part of the **excretory system**

Levels of Organisation

The way in which organisms are built up can be summarised like this:

cells → tissues → organs → organ systems → organism

Level	Meaning	Animal example	Plant example
Cell	The smallest unit from which all organisms are made	A cell lining the stomach	A palisade mesophyll cell
Tissue	A group of similar cells working together to perform a particular function	The layer of cells lining the stomach; muscle tissue	Palisade tissue; epidermis
Organ	A group of different tissues working together to perform specific functions	Stomach, heart, kidney, lungs	Leaf, onion bulb
Organ system	A group of organs with related functions working together	Digestive system (mouth, stomach, intestines)	The shoot system (stem, leaves and flowers)
Organism	A complete living thing	A human	A mango tree

Exam tip: a tissue is made of *similar* cells, while an organ is made of *different* tissues. Mixing these up is a very common mistake.

B1.04 Sizes of specimens

Why We Need Magnification

Many of the structures that biologists study are very small. Cells, for example, are so small that we cannot see them without a microscope. The photographs and diagrams of cells in this chapter are all much larger than the actual cells.

We can tell someone how much bigger an image is than the actual object by giving its **magnification**. The magnification of an image is how many times larger it is than the real object.

The Magnification Equation

The key equation in this chapter is:

magnification = size of image ÷ size of actual object

It can be rearranged to find either of the other two quantities:

- **actual size = size of image ÷ magnification**
- **size of image = actual size × magnification**

A memory aid is a triangle with *image size* in the top half and *actual size* and *magnification* side by side in the bottom half. Cover up the quantity you want: if the two that are left are side by side, multiply them; if one is above the other, divide the top one by the bottom one.

Two Very Important Rules

1. **Make sure that all the numbers in your calculation have the same units.** It is often a good idea to convert everything to millimetres (mm) before you do anything else.
2. **Magnification is always written with a multiplication sign in front of it (×), and it has no units.** For example, write ×500, never 500 mm.

Millimetres and Micrometres

Some of the objects that we study in biology are so small that even millimetres are not a suitable unit for measuring them. Instead, we use **micrometres**. The symbol for a micrometre is **µm**.

- 1 mm = 1000 µm
- 1 µm = 1×10^{-6} m (one millionth of a metre)
- 1 m = 10^{6} µm (one million micrometres)

Conversion	What to do	Example
cm to mm	Multiply by 10	8 cm = 80 mm
mm to µm	Multiply by 1000	2.5 mm = 2500 µm
µm to mm	Divide by 1000	40 µm = 0.04 mm
m to µm	Multiply by 1 000 000	0.002 m = 2000 µm

Rule of thumb: changing to a *smaller* unit makes the number *bigger* (multiply); changing to a *bigger* unit makes the number *smaller* (divide).

Typical sizes: most plant and animal cells are about 10 to 100 µm across; bacteria and mitochondria are only a few µm long. If your answer for the size of a cell comes out as 50 000 µm (5 cm), check your working.

Worked Examples

Worked Example 1: Calculating Magnification (Same Units)

A drawing of an eraser is 120 mm long. The real eraser is 40 mm long. What is the magnification of the drawing?

1. Equation: magnification = size of image ÷ size of actual object
2. Both lengths are already in mm, so no conversion is needed
3. magnification = 120 ÷ 40 = 3
4. Answer: **×3**

Worked Example 2: Converting Units First

A photomicrograph shows a cheek cell that is 30 mm wide. The real cell is 60 µm wide. Calculate the magnification.

1. The units are different, so convert the image size to µm: 30 mm × 1000 = 30 000 µm
2. magnification = 30 000 µm ÷ 60 µm = 500
3. Answer: **×500**

Check by working in mm instead: 60 µm ÷ 1000 = 0.06 mm, and 30 ÷ 0.06 = 500. You get the same answer whichever unit you use, as long as both sizes are in the same unit.

Worked Example 3: Calculating the Actual Size (Question B1.09)

The mitochondrion in Figure B1.10 measures about 83 mm long on the page, and the magnification is ×20 000. What is its real length?

1. Convert the image size to µm: 83 mm × 1000 = 83 000 µm
2. actual size = size of image ÷ magnification = 83 000 µm ÷ 20 000
3. Answer: **4.15 µm**

Worked Example 4: Calculating the Image Size

A cell that is 25 µm wide is drawn at a magnification of ×2000. How wide is the drawing, in mm?

1. size of image = actual size × magnification = 25 µm × 2000 = 50 000 µm
2. Convert to mm: 50 000 µm ÷ 1000 = 50 mm
3. Answer: **50 mm**

Worked Example 5: Actual Size from a Diagram in cm (Practice Question 4)

A diagram of a flower is 8 cm across, and the magnification is ×4.

1. actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm
2. Convert to mm: 2 cm × 10 = 20 mm
3. Answer: **20 mm**

Worked Example 6: Magnification of a Photograph (Practice Question 5)

The jellyfish in the photograph is 89 mm across, and the real jellyfish is 50 mm across.

1. Both sizes are in mm, so magnification = 89 ÷ 50 = 1.78
2. Answer: **×1.8** (to 2 significant figures)

Common Mistakes in Magnification Questions

- **Mixing units:** dividing a length in mm by a length in µm gives an answer that is 1000 times too big or too small. Convert first.
- **Turning the equation upside down:** for an image that has been enlarged, the magnification must be bigger than 1. If you get a number smaller than 1, you have probably divided the actual size by the image size.
- **Multiplying instead of dividing:** to find the actual size you divide the image size by the magnification. Multiplying 8 cm by 4 to get 32 cm is the trap in Practice Question 4.
- **Adding units to a magnification:** ×500 is correct; ×500 mm is wrong.
- **Forgetting the × sign** in front of the magnification.
- **Measuring in cm and calling it mm:** check which marks on your ruler you are reading.

How to Measure an Image

1. Use a ruler marked in millimetres.
2. Measure the length the question asks for (for example, the maximum length or the diameter), in a straight line from one edge of the object to the other.
3. Line up the zero mark of the ruler, not the end of the ruler, with the edge of the object.
4. Record the measurement to the nearest millimetre, and write it down in your working; examiners give a mark for it.

Activity B1.03: Practising Using the Magnification Equation

Aim: To practise measuring objects and drawings, and using the magnification equation.

How it is organised: You work on your own for the first part of the activity, and then pair up with someone else for the second part.

You will need:

- Several small objects that are easy to measure and draw, for example an eraser, a paper clip and a button
- A ruler that can measure in mm
- Some plain paper, a sharp pencil and an eraser

Method, Part 1 (on your own): Step 1 is to measure each object carefully and write down the measurements. Do not let anyone else see these measurements at this point. The rest of the method continues on the next page of the book. In outline, you draw the objects on plain paper, and in the second part you and your partner use the magnification equation with each other's drawings, while your secret measurements are used to check the answers.

How the equation is used: Suppose a paper clip is 32 mm long and your drawing of it is 96 mm long. The magnification of your drawing is 96 ÷ 32 = ×3. If your partner is told that the drawing is ×3 and measures it as 96 mm, they can calculate the real length: 96 ÷ 3 = 32 mm, which matches your secret measurement.

Expected results: If everyone measures carefully in mm, the calculated sizes agree with the real measurements to within about a millimetre. Small differences come from measuring errors.

Conclusion: The magnification equation can be rearranged to find the magnification, the actual size or the image size, as long as all measurements are in the same units.

Safety: There are no special hazards; take care with sharp pencils.

Questions B1.08 and B1.09 (converting units and calculating the real size of a mitochondrion) are answered in the Textbook Questions with Answers section.

Key Formulas and Concepts

1. **Magnification:** magnification = size of image ÷ size of actual object
2. **Actual size:** actual size = size of image ÷ magnification
3. **Image size:** size of image = actual size × magnification
4. **Same units first:** convert both sizes to the same unit (usually mm or μm) before dividing
5. **Millimetres and micrometres:** 1 mm = 1000 μm, so mm × 1000 = μm and μm ÷ 1000 = mm
6. **Centimetres:** 1 cm = 10 mm
7. **Metres:** 1 m = 1000 mm = 1 000 000 μm (10^6 μm), and 1 μm = 1 × 10^{-6} m
8. **Writing magnification:** always with a × sign and no units, for example ×1500
9. **Fitting objects end to end:** number that fit = total length ÷ length of one object, both in the same unit (for example 1000 μm ÷ 4.15 μm ≈ 241 mitochondria in 1 mm)
10. **Microscope power:** a very good light microscope magnifies about ×1500; electron microscopes magnify far more (Figure B1.10 is ×20 000)
11. **Levels of organisation:** cells → tissues → organs → organ systems → organism
12. **Seven characteristics (MRS GREN):** movement, respiration, sensitivity, growth, reproduction, excretion, nutrition
13. **Aerobic respiration (in mitochondria, details in Chapter B9):** glucose + oxygen → carbon dioxide + water, releasing energy
14. **Plant cell extras:** plant cell = animal cell structures + cell wall + large permanent vacuole + (often) chloroplasts
15. **Bacterial cell:** cell wall (not cellulose) + cell membrane + cytoplasm + ribosomes + circular DNA + (often) plasmids; no nucleus, no mitochondria, no chloroplasts

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Exam Focus Priorities

1. **Learn MRS GREN with definitions** — movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. Questions often ask which feature is NOT a characteristic of all organisms: photosynthesis, breathing, a blood system and a nervous system are not.
2. **Respiration is not breathing** — respiration is a set of chemical reactions in every living cell that releases energy from glucose; breathing is moving air in and out of the lungs, which plants and bacteria do not do.
3. **Growth must be permanent** — define growth as a permanent increase in size and dry mass; the word permanent earns the mark.
4. **Know which structures are in which cells** — animal and plant cells both have a cell membrane, cytoplasm, a nucleus, ribosomes and mitochondria; only plant cells have a cellulose cell wall, a large permanent vacuole with cell sap and (often) chloroplasts and starch grains.
5. **Bacteria have no nucleus** — their DNA is a single circle in the cytoplasm, often with plasmids; their cell wall is not made of cellulose; they have no mitochondria or chloroplasts.
6. **Partially permeable versus fully permeable** — the cell membrane is partially permeable and controls what enters and leaves the cell; the cell wall is fully permeable, supports the cell and stops it bursting, but does not control what enters.
7. **Mitochondria release energy** — write that mitochondria release energy from glucose by aerobic respiration; never write that they make or produce energy.
8. **Ribosomes make proteins; DNA decides which proteins** — the DNA in the nucleus carries the instructions, and the ribosomes link amino acids together to build the protein.
9. **No chloroplasts does not mean animal cell** — many plant cells, such as root hair cells, have no chloroplasts; the cell wall is the reliable sign of a plant cell.
10. **Cell, tissue, organ, organ system, organism** — a tissue is a group of similar cells; an organ is made of different tissues; be ready to give an example at each level (stomach lining cell, stomach lining tissue, stomach, digestive system, human).
11. **Show the magnification equation and substitution** — write magnification = image size ÷ actual size, convert to the same units, substitute, then give the answer with a × sign and no units; each step can earn a mark.
12. **Divide to find the actual size** — actual size = image size ÷ magnification; multiplying instead of dividing is the most common error in multiple-choice questions.
13. **Convert mm and µm correctly** — multiply by 1000 to go from mm to µm, divide by 1000 to go from µm to mm, and check that your final size is sensible for the structure.
14. **Measure carefully and record the measurement** — measure in mm with a ruler along the length asked for, and write the measurement in your working.
15. **Use the marks as a guide** — a [6] question needs six separate creditworthy points, and a [2] definition usually needs two ideas.

Textbook Questions with Answers

Question B1.01

A student claimed that plants show fewer of the characteristics of living things than animals. Explain why this claim is wrong.

Answer: The claim is wrong because plants, like all living organisms, show all seven characteristics of living things. Plants simply carry out some of them more slowly or less obviously than animals, which may be why the student was misled.

- **Movement:** plants cannot move their whole body from place to place, but they slowly move parts of themselves; for example, many flowers open during the day and close at night.

- **Respiration:** every living plant cell respires all the time, day and night, releasing energy from glucose.
- **Sensitivity:** plants detect and respond to changes in their environment; for example, shoots grow towards light.
- **Growth:** plants grow throughout their lives, with a permanent increase in size and dry mass.
- **Reproduction:** plants make new plants of the same species, for example from seeds.
- **Excretion:** plants get rid of waste products of metabolism, such as oxygen from photosynthesis and carbon dioxide from respiration.
- **Nutrition:** plants make their own food by photosynthesis and take in water, carbon dioxide and mineral ions.

Question B1.02

Consider *two* organisms. For each, identify:

a) the characteristics of living things that the organism carries out all the time

b) the characteristics of living things that only happen at certain times.

Answer: Answers depend on the organisms you choose. This example uses a human and a mango tree.

a) Human: respiration (every cell respires all the time), excretion (carbon dioxide leaves in every breath out, and the kidneys make urine continuously) and sensitivity (the body is always detecting changes, such as in blood temperature). Mango tree: respiration (all its living cells, day and night), excretion (waste gases leave through the leaves) and sensitivity (it is always detecting light and gravity).

b) Human: growth (mainly until adulthood), reproduction (only adults, and only occasionally), nutrition (we take in food at mealtimes) and movement of the whole body (only when we choose to move). Mango tree: nutrition by photosynthesis (only in daylight), growth (mainly in the growing season), reproduction (flowers and fruit only at certain times of year) and movement (slow movements of leaves and flowers).

Question B1.03

Explain the difference between a unicellular organism and a multicellular organism, giving *one* example of each.

Answer: A unicellular organism is made of just one cell, which carries out all the characteristics of life by itself; an example is a bacterium (or yeast). A multicellular organism is made of many cells (large organisms contain millions of cells), and different cells can become specialised to carry out particular functions; an example is a human (or a plant such as a mango tree).

Question B1.04

Look at the micrographs in Figures B1.05 and B1.06. Suggest why a stain has been added to the animal cells, but we can easily see the plant cells without using a stain.

Answer: Animal cells do not have any coloured parts: the cytoplasm is a clear jelly and the cell membrane is very thin, so the cells and their structures are very difficult to see. The stain colours the cells and makes the nucleus look darker than the cytoplasm, so the structures show up. The moss plant cells contain green chloroplasts, which are already coloured, and their cell walls stand out clearly because they are much thicker than the animal cell membranes. So the plant cells can be seen easily without a stain.

Question B1.05

Name the part of a cell which has each of these functions:

a) makes proteins

b) contains the information for which proteins to make

c) photosynthesis

d) prevents the cell bursting when it takes up water

e) stores a solution of sugars and other solutes

f) controls what enters and leaves the cell.

Answer:

a) Ribosome

b) Nucleus (the DNA in its chromosomes carries the information)

c) Chloroplast

d) Cell wall

e) Vacuole (the large permanent vacuole of a plant cell, which contains cell sap)

f) Cell membrane

Question B1.06

Which of the structures that you have named in Question B1.05 are found in both animal and plant cells?

Answer: The ribosome, the nucleus and the cell membrane are found in both animal and plant cells. The chloroplast and the cell wall are found only in plant cells. A large permanent vacuole containing cell sap is also a plant-cell feature; animal cells only have small vacuoles called vesicles.

Question B1.07

Construct a table to compare the structure of a bacterial cell with animal and plant cells. Remember to include similarities as well as differences.

Answer:

Feature	Bacterial cell	Animal cell	Plant cell
Cell membrane	Yes (partially permeable, pressed against the cell wall)	Yes	Yes
Cell wall	Yes, but not made of cellulose	No	Yes, made of cellulose
Cytoplasm	Yes	Yes	Yes
Ribosomes	Yes	Yes	Yes
Nucleus	No	Yes	Yes
DNA	A single circle of DNA in the cytoplasm	In chromosomes inside the nucleus (not circular)	In chromosomes inside the nucleus (not circular)
Plasmids	Often present	No	No
Mitochondria	No	Yes	Yes
Chloroplasts	No	No	Often, in the green parts of the plant
Large permanent vacuole containing cell sap	No	No (only small vesicles)	Yes

Similarities: all three types of cell have a cell membrane, cytoplasm, ribosomes and DNA; bacterial and plant cells both have a cell wall that supports the cell and stops it bursting. **Main difference:** bacterial cells have no nucleus (they are prokaryotic), and their DNA is circular.

Question B1.08

How many micrometres are there in 1 mm?

Answer: There are 1000 μm in 1 mm. To convert millimetres to micrometres, multiply by 1000.

Question B1.09

The mitochondrion in Figure B1.10 is magnified 20 000 times. (Figure B1.10 is an electron micrograph of one long, oval mitochondrion that stretches almost the whole width of the picture.)

a) Using a ruler, carefully measure the maximum length of the mitochondrion. Record your measurement in mm (millimetres).

b) Convert your answer to μm (micrometres).

c) Use this formula to calculate the real size of the mitochondrion in μm: Real size in μm = size of image in μm ÷ magnification

d) How many of these mitochondria could you line up end to end between two of the mm marks on your ruler?

Answer:

a) About 83 mm, measured in a straight line from one end of the mitochondrion to the other along its longest axis. If your own careful measurement is a millimetre or two different, use your value in parts b to d.

b) 83 mm × 1000 = 83 000 μm

c) Real size = 83 000 μm ÷ 20 000 = 4.15 μm

d) The gap between two mm marks is 1 mm = 1000 μm. Number of mitochondria = 1000 μm ÷ 4.15 μm = 240.96, so about 241 mitochondria (240 whole ones would fit completely).

Practice Exam Questions with Answers

Practice Question 1

Identify the characteristic which is *not* shown by all living organisms. (1 mark)

A) excretion

B) movement

C) photosynthesis

D) respiration

Answer: C — photosynthesis. Only plants (and some other organisms such as algae) photosynthesise; animals, fungi and most bacteria do not. Excretion, movement and respiration are three of the seven characteristics shown by all living organisms.

Practice Question 2

The tongue-eating louse is a parasite of large fish. As its name suggests, it has an unusual way of gaining nutrition. It detects chemicals on the gills of a fish. From the gills it crawls into the fish's mouth. First, the louse sucks blood from the tongue. This causes the tongue to shrink and eventually fall off. Next, the louse takes the place of the tongue. Here the louse remains, feeding on food that the fish takes into its mouth. (7 marks)

a) Define *nutrition*. [2]

b) Apart from nutrition, explain how *two* features of the louse are characteristics of all living organisms. [2]

c) The tongue-eating louse can reproduce.

i) Define *reproduction*. [2]

ii) Suggest *one* reason why scientists know very little about the process of reproduction in this species. [1]

Answer:

a) Taking in substances (materials) from the environment [1]; and using them for energy, growth and development [1].

b) Sensitivity: the louse detects chemicals on the gills of a fish, so it detects and responds to a change in its environment [1]. Movement: the louse crawls from the gills into the fish's mouth, so it changes its position [1].

c) i) The processes that make new organisms [1]; of the same species (the same kind of organism) [1].

c) ii) Any one sensible reason [1], for example: it lives hidden inside the mouths of fish in the sea, so it is very hard to observe; it spends very little time outside its host; it is rare in the wild; it is difficult to keep and breed in a laboratory.

Practice Question 3

The diagram below shows a cell from a plant leaf. Identify which labelled part contains DNA. (1 mark)

A) the outer double line that forms the boundary of the cell

B) the small oval structure at the left-hand side of the cell

C) the narrow layer between the outer boundary and the large central space (top right)

D) the large central space that fills most of the cell

Answer: B — the small oval structure is the nucleus, which contains DNA in the form of chromosomes. (A is the cell wall, C is the cytoplasm and D is the large permanent vacuole.)

Practice Question 4

A diagram of a flower is 8 cm across. The magnification of the diagram is ×4. Identify the actual width of the flower. (1 mark)

A) 3.2 mm

B) 20 mm

C) 0.5 cm

D) 32 cm

Answer: B — 20 mm. Actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm, and 2 cm × 10 = 20 mm. D (32 cm) comes from multiplying instead of dividing, and C (0.5 cm) comes from dividing 4 by 8.

Practice Question 5

The photograph shows a jellyfish. A jellyfish belongs to the animal kingdom. (9 marks)

(The photograph is a black-and-white side view of a jellyfish: a see-through, dome-shaped bell with a frilly edge and short fringes hanging underneath. Measured across its widest part, the jellyfish in the printed photograph is 89 mm wide.)

a) The actual jellyfish is 50 mm in diameter. **Calculate** the magnification of the photograph. [3]

b) State *three* parts that you would find in a cell in a jellyfish *and* outline the function of each part. [6]

Answer:

a) Magnification = size of image ÷ size of actual object [1]; = 89 mm ÷ 50 mm [1]; = 1.78, so the magnification is ×1.8 to 2 significant figures [1]. The answer must have a × sign and no units. If your own measurement of the photograph is slightly different, use it in the same way.

b) Any three parts, each with its function (1 mark for the part and 1 mark for its function):

- Nucleus [1]: contains DNA in the form of chromosomes, carrying the information about which proteins the cell makes [1]
- Cell membrane [1]: partially permeable; controls what enters and leaves the cell [1]
- Cytoplasm [1]: the jelly where many metabolic reactions take place [1]
- Mitochondrion [1]: the site of aerobic respiration, which releases energy from glucose [1]
- Ribosome [1]: where proteins are made [1]

Do not give a cell wall, chloroplasts or a large vacuole containing cell sap: a jellyfish is an animal, so its cells do not have these.

Practice Question 6

This is an electron micrograph of a small part of a cell from the pancreas. This cell makes large quantities of protein molecules, which are stored in vesicles before being exported from the cell. (8 marks)

(In the micrograph, label A points to a large, pale, finely grainy area at the top left. Label B points to a long, pale, sausage-shaped structure in the centre with fine stripes across its inside. Label C points to the dark, closely packed wavy lines that fill most of the background. Several round, dark grey structures are labelled 'vesicle containing protein molecules made by the cell'.)

a) Structure A contains molecules that determine the type of proteins made in the cell. Identify structure A. [1]

b) Use the information above to explain why the cell has large numbers of structures B and C. [6]

c) State *one* way you can tell that these are not bacterial cells. [1]

Answer:

a) The nucleus [1].

b) Any six from:

- B is a mitochondrion [1]
- Mitochondria are the site of aerobic respiration [1]
- Aerobic respiration releases energy from glucose [1] (do not write that mitochondria produce energy)
- Making (synthesising) proteins requires energy [1], and so does moving the vesicles and exporting the proteins from the cell
- C shows ribosomes; the dark, grainy lines are membranes covered in ribosomes [1]
- Ribosomes are the site of protein synthesis, where amino acids are linked together to make proteins [1]
- The cell makes large quantities of protein, so it needs many ribosomes to make them and many mitochondria to supply the energy [1]

c) Any one [1]: the cell has a nucleus (A); it has mitochondria (B); it has vesicles. Bacterial cells have no nucleus, no mitochondria and no vesicles.

Summary of Key Concepts

Chapter Summary

- **Seven characteristics:** all organisms show seven characteristics: movement, respiration, sensitivity, growth, reproduction, excretion and nutrition.

- **Cells:** all organisms are made of cells, and new cells are always made by the division of existing cells.
- **Animal and plant cells:** both contain cytoplasm, a cell membrane, ribosomes, mitochondria and a nucleus containing DNA in the form of chromosomes. Plant cells also contain a cell wall, a large vacuole and sometimes chloroplasts.
- **Bacterial cells:** they have cytoplasm, a cell membrane, a cell wall and ribosomes, but no nucleus. Their DNA is circular, and they may have extra, small circles of DNA called plasmids.
- **Membrane and wall:** the cell membrane of all cells is partially permeable and controls what enters and leaves the cell. The cell wall is fully permeable and allows all molecules and ions to pass through it.
- **Ribosomes:** the site of protein synthesis in a cell.
- **Mitochondria:** release energy from glucose and other nutrients by aerobic respiration.
- **Chloroplasts:** the site of photosynthesis.
- **Specialised cells:** cells may be specialised for specific functions. Similar cells form tissues, tissues form organs, organs form organ systems, and organ systems make up the organism.
- **Magnification:** magnification = size of image ÷ size of actual object.
- **Units:** convert millimetres to micrometres by multiplying by 1000 (1 mm = 1000 μm).

Exam Technique Tips

- In magnification questions, always write the equation, convert to the same units, substitute your numbers, and give the answer with a × sign and no units.
- When you measure an image, write your measurement down; it often earns the first mark.
- For "explain" questions about cell structures, link each structure to its function and then to the information in the question (for example, a cell that makes a lot of protein needs many ribosomes and many mitochondria).
- Use precise biological words: partially permeable, aerobic respiration, protein synthesis, and release energy (not produce energy).
- Read the question carefully: "not shown by all organisms" and "found in both animal and plant cells" are easy to misread.
- The mark allocation for a question often tells you how many points you need to make to answer it fully.

Command Words Used in This Chapter

- **Identify:** name/select/recognise.
- **Define:** give a precise meaning.
- **Explain:** set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence.
- **Suggest:** apply knowledge and understanding to situations where there is a range of valid responses in order to make proposals/put forward considerations.
- **Calculate:** work out from given facts, figures or information.
- **State:** express in clear terms.

Self-Check

- I can define the seven characteristics of living organisms (B1.01)
- I can describe how all organisms are made of cells, and how new cells are always made by the division of existing cells (B1.02)
- I can describe and compare the structure of a bacterial cell, an animal cell and a plant cell (B1.02)
- I can describe the functions of each of the structures found in an animal cell and a plant cell (B1.02)
- I can identify cell structures in diagrams and micrographs (B1.02)

- I can name the seven specialised cells in Table B1.01 and state their functions (B1.03)
- I can explain the terms cell, tissue, organ, organ system and organism, with examples (B1.03)
- I can calculate magnification, actual size and image size, showing my working (B1.04)
- I can convert between millimetres and micrometres (B1.04)